

DDP · Dal Degree Planner

Preliminary design brief · September–December 2026 · Backlog due September 27

Company and project information

Client/company: TBD — confirm official organization name. **Representative:** TBD. **Team:** Project Director TBD; Technical Director TBD; two senior and two junior developers (names TBD; six-person roster confirmed). Proposed hosting: Dalhousie FCS servers; capabilities pending confirmation.

Problem and users

BACS students need to understand course dependencies, co-requisites and grade thresholds while creating manageable multi-year plans. Students use DDP without login; designated administrators maintain reviewed course and degree data.

Proposed product and scope

A responsive course explorer, linked official descriptions/timetable, visual dependency map and term planner with drag/drop plus accessible add/move controls. Student plans save in the browser and export/import as JSON. Checks distinguish earned, pending, blocked and unknown requirements. Admin login and validated course/requirement editing need a backend. Live timetable synchronization and comprehensive transfer/exception audits are future scope unless agreed.

Design and architecture

React with plain JavaScript, Tailwind for layout and Material UI for controls. Small components, separate curriculum data and pure rule functions; browser-local student plan. Versioned curriculum API and persistent store for admin, subject to FCS hosting decision. Dal-inspired black/gold styling. Workbook and calendar rules require source/year/reviewer records; the planner does not certify graduation eligibility.

Workload and responsibilities

Role	Planned contribution
Project Director	Client, backlog, coordination, acceptance and reporting; occasional frontend.
Technical Director	Architecture, backend/integration, hosting, reviews and coding.
Two seniors	Main feature coding; frontend and domain ownership; one junior mentee each.
Two juniors	Small UI, source transcription, fixtures, documentation and supported QA.

Initial capacity proposal: 5 hours/person/week, 30 team hours; about 16 for new delivery after coordination, mentoring and contingency. Confirm individually and re-estimate each sprint.

Tools and Git organization

VS Code, Node/npm, React/Vite, Tailwind/MUI, Teams and Git. Proposed GitHub organization/repo: TBD; visibility: TBD. Main holds reviewed work; short feature branches, linked issues and PR review by another developer. CI runs app checks/build using a committed lockfile. Every member pushes evidence before graded deadlines; no credentials committed.

Delivery and acceptance

Sep. 20 setup; Sep. 27 backlog/brief; Oct. 11 usable student slice; Oct. 25 rules/map; Nov. 29 integrated candidate; Dec. 9 packaged executable. Presentation and closing dates TBD. Acceptance: a non-technical tester builds a plan, understands a dependency issue, exports/reimports it, and follows the guide; an authorized admin maintains data. Main risks: unreviewed rules, hosting availability and limited term capacity.